LP3 PRACTICAL VIVA QUESTIONS

Group A: DDA

Practical 1: Fibonacci using recursion and iteration

Q1. What is the Fibonacci series?

A. It is a series in which each number is the sum of the previous two numbers: 0, 1, 1, 2, 3, 5, 8, 13...

Q2. What is the formula?

A. $F(n)=F(n-1)+F(n-2)$, with $F(0)=0$ and $F(1)=1$.

Q3. What is recursion?

A. Recursion is a technique in which a function calls itself to solve a smaller problem.

Q4. What is iteration?

A. Iteration repeats a set of statements using loops such as for or while.

Q5. What is the base case?

A. The condition that stops recursion. Here, if n <= 1, return n.

Q6. What happens if the base case is missing?

A. Recursion may continue until a recursion error or stack overflow occurs.

Q7. What is the time complexity of recursive Fibonacci?

A. $O(2^n)$ for the usual naive recursive implementation.

Q8. What is its space complexity?

A. $O(n)$ because of the maximum recursion stack depth.

Q9. What is the complexity of iterative Fibonacci?

A. $O(n)$ time and $O(1)$ auxiliary space when only two previous values are stored.

Q10. Which method is better and why?

A. Iteration is generally better for this problem because it avoids repeated calculations and uses less memory.

Q11. What is the output for n = 7 terms?

A. 0 1 1 2 3 5 8

Q12. What is the 9th Fibonacci number?

A. If indexing begins at $F(0)=0$, then $F(9)=34$.

Q13. What is the difference between printing n terms and finding the nth term?

A. Printing n terms displays the first n numbers; finding the nth term returns one number at index n.

Q14. What is memoization?

A. It stores previously calculated results so they do not need to be calculated again.

Q15. What is the Golden Ratio?

A. It is approximately \(1.618\), related to the ratio of consecutive Fibonacci numbers as the sequence grows.

Practical 2: Huffman Encoding using a greedy strategy

Q1. What is Huffman Encoding?

A. It is a lossless compression technique that assigns variable-length binary codes according to character frequencies.

Q2. Which algorithmic approach does it use?

A. Greedy approach.

Q3. Who developed Huffman Coding?

A. David A. Huffman.

Q4. Why is it called lossless compression?

A. The original data can be recovered exactly after decoding.

Q5. What is a Huffman tree?

A. A binary tree used to generate codes for characters based on their frequencies.

Q6. What is a prefix code?

A. A code in which no character's code is the prefix of another character's code.

Q7. Why is the prefix property important?

A. It prevents ambiguity during decoding.

Q8. Which character gets the shortest code?

A. Generally, the most frequent character.

Q9. What are the main steps?

A. Count frequencies, place nodes in a min-priority queue, repeatedly combine the two minimum-frequency nodes, construct the tree, and assign binary codes.

Q10. What do 0 and 1 represent?

A. They represent the left and right edges of the tree, respectively; the convention can be reversed.

Q11. What is a min-heap?

A. A heap in which the smallest element is at the root.

Q12. What is the time complexity?

A. \(O(n\log n)\) when a min-heap is used for n distinct characters.

Q13. Can two characters have codes of the same length?

A. Yes. Their codes can have equal lengths while remaining distinct and prefix-free.

Q14. Does Huffman Coding always assign equal-length codes?

A. No. It assigns variable-length codes.

Q15. What happens when two characters have equal frequencies?

A. Either can be selected first; different valid trees and codes may result, but the optimal encoded cost can remain the same.

Q16. How is the encoded message length calculated?

A. Multiply each character's frequency by its code length and add the results.

$$ L=\sum_{i=1}^{n} f_i\,l_i $$

Here, f_i is frequency and l_i is code length.

Practical 3: Fractional Knapsack using Greedy Method

Q1. What is the Knapsack problem?

A. It is an optimization problem in which we select items to maximize total value without exceeding a capacity limit.

Q2. What is Fractional Knapsack?

A. A problem where items can be divided, so a fraction of an item can be selected.

Q3. Which approach is used?

A. Greedy approach.

Q4. What is the selection criterion?

A. Select items in decreasing order of their value-to-weight ratio.

$$ \text{Ratio}=\frac{\text{Value}}{\text{Weight}} $$

Q5. Why do we calculate this ratio?

A. It tells us how much value we get per unit of weight.

Q6. Can we take half an item?

A. Yes, in Fractional Knapsack.

Q7. What is the difference between Fractional and 0/1 Knapsack?

A. Fractional Knapsack allows partial items; 0/1 Knapsack allows only complete selection or rejection.

Q8. What is the time complexity?

A. $O(n\log n)$, mainly because of sorting the items.

Q9. What if the knapsack has no remaining capacity?

A. Stop selecting further items.

Q10. What if an item's weight exceeds the remaining capacity?

A. Take only the fraction that fits, then stop.

Q11. Why does the greedy method work for Fractional Knapsack?

A. Because choosing the highest value per unit of weight first maximizes the total value when fractions are allowed.

Q12. Does the same greedy strategy always solve 0/1 Knapsack optimally?

A. No. Items cannot be divided in 0/1 Knapsack, so choosing the highest ratio first may miss the optimal combination.

Q13. What is the total value if an item worth ₹100 weighs 20 kg and we take 5 kg?

A. $100\times5/20=25$ units of value.

Practical 4: 0/1 Knapsack using Dynamic Programming

Q1. What is Dynamic Programming (DP)?

A. It solves a problem by storing solutions to subproblems and reusing them.

Q2. What are the two main properties of DP?

A. Overlapping subproblems and optimal substructure.

Q3. What is 0/1 Knapsack?

A. A problem where each item is either selected completely or not selected at all.

Q4. What does dp[i][w] represent?

A. The maximum value obtainable using the first i items with capacity w.

Q5. What is the recurrence relation?

A. If the item fits:

$$ dp[i][w]=\max(dp[i-1][w],v_i+dp[i-1][w-w_i]) $$

Otherwise, $dp[i][w]=dp[i-1][w]$.

Q6. Why is the first row initialized to zero?

A. With zero items, the maximum value is zero.

Q7. Why is the first column initialized to zero?

A. With zero capacity, no item can be selected.

Q8. What is the time complexity?

A. $O(nW)$, where n is the number of items and W is the capacity.

Q9. What is the space complexity of the full DP table?

A. $O(nW)$.

Q10. Can we optimize the space?

A. Yes, using a one-dimensional DP array, reducing auxiliary space to $O(W)$.

Q11. Why does the one-dimensional implementation loop backward over capacity?

A. To avoid using an item more than once in the same iteration.

Q12. What is Branch and Bound?

A. An optimization technique that explores candidate solutions while using bounds to prune branches that cannot improve the best solution.

Q13. How is DP different from Branch and Bound?

A. DP stores subproblem solutions; Branch and Bound prunes unpromising search branches using bounds.

Q14. Is \(O(nW)\) polynomial time in the input size?

A. It is pseudo-polynomial because the complexity depends on the numeric value of W, not just the number of bits needed to represent W.

Practical 5: N-Queens using Backtracking

Q1. What is the N-Queens problem?

A. Place N queens on an \(N\times N\) chessboard so no two queens attack each other.

Q2. What is backtracking?

A. It tries a choice, checks whether it can lead to a solution, and reverses the choice if it fails.

Q3. How can a queen attack another queen?

A. Horizontally, vertically, or diagonally.

Q4. Why is recursion used?

A. To place queens one at a time and solve the remaining smaller problem.

Q5. What is the base case?

A. When all N queens have been placed safely, a solution is found.

Q6. Why do we remove a queen during backtracking?

A. To try a different position when the current placement leads to a dead end.

Q7. Why is it sufficient to check only previously filled rows or columns?

A. Because the algorithm places one queen per row or column in sequence, so unfilled rows or columns cannot yet contain attacking queens.

Q8. Is a solution possible for N = 2 or N = 3?

A. No. Neither board size has a solution.

Q9. Give one solution for 4-Queens.

A. The row positions by column can be represented as (2, 4, 1, 3).

Q10. What does a 1 and a 0 represent in the matrix?

A. 1 represents a queen; 0 represents an empty cell.

Q11. What is the worst-case time complexity?

A. It is exponential; a common upper bound for the straightforward backtracking approach is \(O(N!)\).

Q12. Name other applications of backtracking.

A. Sudoku, maze solving, graph coloring, and the sum-of-subsets problem.

Q13. Does backtracking always find every solution?

A. It can enumerate all solutions if implemented to continue searching after finding one; a version that returns immediately may find only one.

Group B — Machine Learning

A. Basic Machine Learning viva questions

Q1. What is Machine Learning?

A. Machine Learning is a branch of AI that enables computers to learn patterns from data and make predictions or decisions.

Q2. What are the main types of Machine Learning?

A. Supervised learning, unsupervised learning, and reinforcement learning.

Q3. What is supervised learning?

A. Learning from labelled data, where the correct output is known.

Q4. What is unsupervised learning?

A. Finding patterns or groups in unlabelled data.

Q5. What is reinforcement learning?

A. Learning through rewards and penalties based on actions.

Q6. What is the difference between classification and regression?

A. Classification predicts categories; regression predicts continuous numerical values.

Q7. Give examples of classification.

A. Spam detection, diabetes prediction, and customer churn prediction.

Q8. Give examples of regression.

A. Predicting house prices, taxi fares, or temperature.

Q9. What is a dataset?

A. A collection of data used for analysis and model training.

Q10. What are features and target variables?

A. Features are input variables; the target is the output the model predicts.

Q11. What is training data?

A. Data used to teach the model patterns.

Q12. What is testing data?

A. Data used to evaluate the model on unseen examples.

Q13. What is overfitting?

A. When a model learns training data too closely and performs poorly on unseen data.

Q14. What is underfitting?

A. When a model is too simple to learn the underlying patterns.

Q15. What is generalization?

A. The ability of a model to perform well on new, unseen data.

Q16. What is data preprocessing?

A. Cleaning and transforming raw data into a suitable form for machine learning.

Q17. Why is preprocessing necessary?

A. To handle missing values, incorrect data, categorical variables, inconsistent scales, and outliers.

Q18. What is feature selection?

A. Choosing the most relevant input variables for a model.

Q19. What is feature scaling?

A. Adjusting numerical features to comparable scales.

Q20. What is a random state or random seed?

A. A fixed value used to make randomized operations reproducible.

Q21. What is cross-validation?

A. A technique that evaluates a model using multiple training and validation splits.

Q22. What is a hyperparameter?

A. A setting chosen before or during model tuning, such as K in KNN or tree depth.

Q23. What is a model?

A. A learned mathematical or computational representation of patterns in data.

Q24. What is scikit-learn?

A. A Python library for machine learning, preprocessing, model selection, and evaluation.

Q25. Why do we use Pandas?

A. To load, clean, manipulate, and analyze tabular data.

Q26. Why do we use NumPy?

A. For efficient numerical calculations and array operations.

Q27. Why do we use Matplotlib and Seaborn?

A. To create visualizations and understand data patterns.

Q28. What is a CSV file?

A. A Comma-Separated Values file used to store tabular data.

B. Data preprocessing and evaluation

Q29. How do you read a CSV file in Python?

A. pd.read_csv("file.csv")

Q30. How do you display the first five rows?

A. df.head()

Q31. How do you check the number of rows and columns?

A. df.shape

Q32. How do you check missing values?

A. df.isnull().sum()

Q33. How do you remove duplicate rows?

A. df.drop_duplicates()

Q34. What are missing values?

A. Values absent from the dataset, often represented as NaN or null.

Q35. How can missing values be handled?

A. By removing affected rows or columns, or imputing values using methods such as mean, median, or mode.

Q36. What is categorical encoding?

A. Converting categories into numerical representations that a model can process.

Q37. What is label encoding?

A. Assigning an integer to each category.

Q38. What is one-hot encoding?

A. Creating separate binary columns for categories.

Q39. What is normalization?

A. Scaling data, often to the range 0 to 1.

$$ X'=\frac{X-X_{\min}}{X_{\max}-X_{\min}} $$

Q40. What is standardization?

A. Transforming values to have a mean of approximately 0 and standard deviation of 1.

$$ Z=\frac{X-\mu}{\sigma} $$

Q41. What is the difference between normalization and standardization?

A. Min-max normalization usually scales values to a fixed range; standardization uses the mean and standard deviation and has no fixed range.

Q42. Why is scaling important for KNN and K-Means?

A. Both depend on distances, so features with larger numerical ranges could otherwise dominate.

Q43. Does every algorithm require scaling?

A. No. Distance-based algorithms commonly benefit from it; tree-based models generally do not require it.

Q44. What is an outlier?

A. A value that differs substantially from most observations.

Q45. How can outliers be detected?

A. Using box plots, the IQR method, Z-scores, or domain knowledge.

Q46. What is the IQR?

A. The interquartile range, calculated as Q_3-Q_1.

Q47. What is correlation?

A. A measure of the strength and direction of the relationship between two variables.

Q48. What is Pearson correlation?

A. A measure of the strength and direction of a linear relationship, ranging from −1 to +1.

Q49. Does correlation prove causation?

A. No. Correlation alone does not establish that one variable causes another.

Q50. Why do we split the dataset?

A. To train the model on one portion and evaluate its performance on unseen data.

Q51. What does train_test_split() do?

A. It divides arrays or datasets into training and testing subsets.

Q52. What does test_size=0.2 mean?

A. 20% of the data is allocated to the test set.

Q53. What is data leakage?

A. When information that would not legitimately be available during training influences the model, producing misleading evaluation results.

Q54. How can leakage be prevented?

A. Split the data first and fit preprocessing steps only on training data, preferably using a pipeline.

C. Regression and Uber fare prediction

Q55. What is Linear Regression?

A. A supervised learning algorithm that models a continuous target using a linear relationship with input features.

Q56. What is the equation of simple Linear Regression?

A. $y=mx+c$, where m is the slope and c is the intercept.

Q57. What is Multiple Linear Regression?

A. Linear regression with multiple input features.

Q58. What is Random Forest Regression?

A. A model that combines predictions from multiple decision trees, typically by averaging their outputs.

Q59. What is the difference between Linear Regression and Random Forest Regression?

A. Linear Regression fits a linear relationship; Random Forest can capture more complex nonlinear patterns.

Q60. What is a residual?

A. The difference between an actual value and its predicted value.

Q61. What is MSE?

A. Mean Squared Error, the average of squared prediction errors.

$$ MSE=\frac{1}{n}\sum_{i=1}^{n}(y_i-\hat y_i)^2 $$

Q62. What is RMSE?

A. The square root of MSE. It is expressed in the same units as the target variable.

Q63. What is R^2 score?

A. A measure of how well a regression model explains variation in the target compared with a baseline prediction.

Q64. Is a higher or lower RMSE better?

A. Lower RMSE is generally better when comparing models on the same test set.

Q65. Is a higher or lower R^2 better?

A. Generally, higher is better; 1 indicates perfect predictions, and the score can be negative.

Q66. What is Haversine distance?

A. A formula for calculating the great-circle distance between two points on a sphere using their latitude and longitude.

Q67. Why is Haversine used in Uber fare prediction?

A. It estimates the geographic distance between pickup and drop-off locations.

Q68. What is a box plot used for?

A. To display the median, quartiles, spread, and potential outliers.

Q69. What is the target variable in fare prediction?

A. The fare amount.

Q70. Why can invalid latitude or longitude values affect predictions?

A. They can produce unrealistic distances and distort the learned relationship between location and fare.

D. KNN and diabetes prediction

Q71. What is K-Nearest Neighbors (KNN)?

A. A supervised learning algorithm that predicts a label based on the labels of nearby training examples.

Q72. How does KNN work?

A. It calculates distances, selects the K nearest data points, and uses majority voting for classification.

Q73. What does K represent?

A. The number of nearest neighbors considered.

Q74. What happens when K = 1?

A. The prediction depends on the single nearest training example, making it sensitive to noise.

Q75. What happens when K is too large?

A. The model may become less sensitive to local patterns and underfit.

Q76. Why is KNN called a lazy learner?

A. It stores training examples and does most of its computation when a prediction is requested.

Q77. Which distance measure is commonly used in KNN?

A. Euclidean distance.

\[d=\sqrt{\sum_{i=1}^{n}(x_i-y_i)^2} \]

Q78. Why is scaling important for diabetes prediction using KNN?

A. Features such as age, BMI, and glucose can have different scales, affecting distance calculations.

Q79. What is the target column in the diabetes dataset described in the manual?

A. Outcome, representing the diabetes class.

Q80. What are examples of features in this dataset?

A. Pregnancies, BMI, insulin level, and age.

Q81. Why is KNN a classification algorithm here?

A. The target represents categories, such as diabetes or no diabetes.

Q82. Can KNN also be used for regression?

A. Yes. It can predict a numerical value using nearby observations.

E. Confusion matrix and classification metrics

Q83. What is a confusion matrix?

A. A table comparing actual and predicted class labels.

Q84. What are TP, TN, FP, and FN?

A. True Positive, True Negative, False Positive, and False Negative.

Q85. What is accuracy?

A. The proportion of all predictions that are correct.

$$ Accuracy=\frac{TP+TN}{TP+TN+FP+FN} $$

Q86. What is the error rate?

A. The proportion of predictions that are incorrect: $1-Accuracy$.

Q87. What is precision?

A. The proportion of predicted positive cases that are actually positive.

$$ Precision=\frac{TP}{TP+FP} $$

Q88. What is recall?

A. The proportion of actual positive cases correctly identified.

$$ Recall=\frac{TP}{TP+FN} $$

Q89. What is the F1-score?

A. The harmonic mean of precision and recall.

$$ F1=2\frac{Precision\times Recall}{Precision+Recall} $$

Q90. What is a false positive?

A. The model predicts positive when the actual class is negative.

Q91. What is a false negative?

A. The model predicts negative when the actual class is positive.

Q92. Which metric is important when missing a positive case is costly?

A. Recall is particularly important, though precision and other consequences must also be considered.

Q93. What is the ROC curve?

A. A graph of true positive rate against false positive rate at different classification thresholds.

Q94. What is ROC-AUC?

A. The area under the ROC curve, which measures a classifier's ability to distinguish classes.

Q95. What is class imbalance?

A. When one class has many more examples than another.

Q96. Why can accuracy be misleading?

A. A model can achieve high accuracy by mostly predicting the majority class while missing the minority class.

F. Email spam detection using KNN and SVM

Q97. What is binary classification?

A. Classification involving two possible classes.

Q98. What are the two classes in email spam detection?

A. Spam and not spam.

Q99. What is Support Vector Machine (SVM)?

A. A supervised algorithm that finds a decision boundary separating classes, aiming for a large margin.

Q100. What is a support vector?

A. A training point that helps determine the separating boundary.

Q101. What is a hyperplane?

A. A decision boundary that separates classes in the feature space.

Q102. What is the kernel trick?

A. A technique that allows SVM to model nonlinear boundaries by implicitly representing data in a higher-dimensional feature space.

Q103. Name common SVM kernels.

A. Linear, polynomial, RBF, and sigmoid.

Q104. What is the difference between KNN and SVM?

A. KNN predicts from nearby examples; SVM learns a decision boundary.

Q105. Why is text converted into numerical features?

A. Machine-learning algorithms need numerical representations to process text.

Q106. What does a spam dataset with 3,000 word columns represent?

A. Each word column stores a numerical count or feature for that word in an email.

Q107. What is the target in the manual's spam dataset?

A. 1 for spam and 0 for not spam.

Q108. Why can SVM work well with high-dimensional text features?

A. SVM can learn effective decision boundaries in spaces with many features.

G. Artificial Neural Network and bank customer churn

Q109. What is an Artificial Neural Network (ANN)?

A. A machine-learning model made of interconnected computational units called neurons.

Q110. What are the main layers of an ANN?

A. Input layer, hidden layer or layers, and output layer.

Q111. What is a neuron?

A. A computational unit that combines weighted inputs, adds a bias, and applies an activation function.

Q112. What is an activation function?

A. A function that transforms a neuron's input and helps the network learn nonlinear patterns.

Q113. Name common activation functions.

A. ReLU, sigmoid, and tanh.

Q114. What is ReLU?

A. The function $f(x)=\max(0,x)$.

Q115. What is sigmoid?

A. A function that maps real-valued inputs to values between 0 and 1.

Q116. What is backpropagation?

A. A method for calculating gradients of the loss with respect to network parameters.

Q117. What is an epoch?

A. One complete pass through the training dataset.

Q118. What is batch size?

A. The number of training examples processed before a parameter update.

Q119. What is a loss function?

A. A function that measures the difference between predictions and the actual targets.

Q120. What is an optimizer?

A. An algorithm that updates model parameters to reduce the loss.

Q121. What is TensorFlow?

A. An open-source framework for numerical computation and machine learning.

Q122. What is Keras?

A. A high-level API for building and training neural networks.

Q123. What is customer churn?

A. When a customer stops using a company's product or service.

Q124. What is the target in the bank churn practical?

A. Whether a customer leaves or stays, represented by the target column in the dataset.

Q125. Why is normalization used before training the ANN?

A. To put numerical features on comparable scales and help training behave more reliably.

H. K-Means and hierarchical clustering

Q126. What is clustering?

A. Grouping similar data points without requiring predefined labels.

Q127. Is K-Means supervised or unsupervised?

A. Unsupervised.

Q128. What does K represent in K-Means?

A. The number of clusters to create.

Q129. What is a centroid?

A. The center of a cluster, usually calculated as the mean of its assigned points.

Q130. Explain K-Means steps.

A. Initialize centroids, assign points to the nearest centroid, recompute centroids, and repeat until convergence.

Q131. What is the elbow method?

A. A method for choosing K by plotting within-cluster sum of squares against the number of clusters and looking for a bend in the curve.

Q132. What is inertia in K-Means?

A. The sum of squared distances between data points and their assigned centroids.

Q133. What is hierarchical clustering?

A. A clustering method that builds a hierarchy of nested clusters.

Q134. What are agglomerative and divisive clustering?

A. Agglomerative starts with individual points and merges clusters; divisive starts with one cluster and splits it.

Q135. What is a dendrogram?

A. A tree-like diagram showing how clusters merge or split.

Q136. What is the difference between K-Means and hierarchical clustering?

A. K-Means requires K in advance and iteratively updates centroids; hierarchical clustering creates a cluster hierarchy that can be cut at a chosen level.

Q137. Why is scaling important for customer segmentation?

A. Features such as income and spending score can have different ranges and otherwise influence distance disproportionately.

Q138. What is the objective of customer segmentation?

A. To identify groups with similar characteristics or purchasing behaviour.

I. Practical, code and tricky questions

Q139. What does model.fit(X_train, y_train) do?

A. It trains the model using the training features and labels.

Q140. What does model.predict(X_test) do?

A. It predicts outputs for the test features.

Q141. What does model.score() return?

A. It depends on the estimator: commonly accuracy for classifiers and \(R^2\) for regressors.

Q142. What is a pipeline?

A. A sequence of preprocessing and model steps treated as one workflow.

Q143. What if your model gives very high training accuracy but low testing accuracy?

A. It may be overfitting.

Q144. What if your dataset contains text categories but the model expects numbers?

A. Encode the categorical features appropriately.

Q145. Why should we not use the test set repeatedly to tune a model?

A. Repeated tuning on the test set makes the evaluation biased. Use validation data or cross-validation for tuning.

Q146. Can a model with 99% accuracy always be trusted?

A. No. Check class balance, data leakage, test-set quality, and other evaluation metrics.

Q147. Why do we compare multiple algorithms?

A. To identify which approach performs best for the particular dataset and objective.

Q148. What is reproducibility in a machine-learning experiment?

A. Getting consistent results when repeating the experiment under the same conditions.

Group C — Blockchain Technology

Your manual includes blockchain fundamentals, Ethereum, MetaMask, Ether transactions, smart contracts, Solidity, and Hyperledger-related practical work. It also includes a survey of blockchain types and their real-world uses.

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A. Blockchain fundamentals

Q1. What is Blockchain?

A. Blockchain is a distributed digital ledger that stores records in linked blocks and makes unauthorized changes difficult.

Q2. Why is it called a blockchain?

A. Because blocks of records are linked together using cryptographic hashes.

Q3. What is a block?

A. A block contains transaction data, metadata, and a reference to the previous block.

Q4. What is a hash?

A. A fixed-length output generated from input data by a hash function.

Q5. What happens if data in a block is changed?

A. Its hash changes, which can make the alteration detectable and break the expected chain of references.

Q6. What is a distributed ledger?

A. A record of transactions shared and maintained across multiple participating computers.

Q7. What is decentralization?

A. Control is distributed across participants rather than held by a single central authority.

Q8. What is a node?

A. A computer or participant that performs some role in a blockchain network.

Q9. What is a transaction?

A. An operation that transfers assets or changes blockchain state.

Q10. What is consensus?

A. A process through which network participants agree on valid transactions or the state of the blockchain.

Q11. What is immutability?

A. The property that recorded information is difficult to change after confirmation, although immutability is not an absolute guarantee against every attack or protocol change.

Q12. Is blockchain completely unhackable?

A. No. Smart-contract bugs, stolen keys, software vulnerabilities, and network attacks can still cause losses.

Q13. What is cryptography?

A. Techniques for protecting information, verifying authenticity, and securing communications or transactions.

Q14. What is a digital signature?

A. A cryptographic mechanism used to verify that a message or transaction was authorized by the holder of a private key.

Q15. What is the difference between a public key and a private key?

A. A public key can be shared; a private key must remain secret and is used to create digital signatures.

Q16. What is a peer-to-peer network?

A. A network in which participants can communicate directly without relying entirely on a central server.

Q17. What is a cryptocurrency?

A. A digital asset that uses cryptographic techniques and may operate on a blockchain or another distributed system.

Q18. Is blockchain the same as cryptocurrency?

A. No. Blockchain is a technology; cryptocurrency is one possible application of it.

Q19. What is a 51% attack?

A. An attack in which an entity gains majority control over a blockchain's relevant consensus resource, potentially enabling certain transaction reorganizations or censorship.

Q20. What are the applications of blockchain?

A. Payments, supply-chain tracking, digital identity, asset tokenization, and shared recordkeeping.

B. Types of blockchain

Q21. What are the four types of blockchain in the manual?

A. Public, private, hybrid, and consortium blockchain.

Q22. What is a public blockchain?

A. A network that generally allows anyone to participate under its protocol rules.

Q23. Give examples of public blockchains.

A. Bitcoin and Ethereum.

Q24. What is a private blockchain?

A. A blockchain with restricted access, typically controlled by one organization.

Q25. What is a consortium blockchain?

A. A blockchain governed by multiple selected organizations.

Q26. What is a hybrid blockchain?

A. A system that combines public-facing and permission-restricted elements.

Q27. What is the difference between public and private blockchain?

A. Public blockchains are generally open to participation; private blockchains restrict participation to authorized users.

Q28. Which blockchain is suitable for multiple banks working together?

A. A permissioned consortium blockchain may be suitable when the banks need shared control and restricted access.

Q29. Which blockchain is more transparent?

A. Public blockchains generally offer broader public visibility, although the visibility of specific data depends on the protocol.

Q30. What is one disadvantage of a public blockchain?

A. It may have limited transaction throughput, fees, and privacy challenges.

Q31. What is one advantage of a private blockchain?

A. It can provide controlled access, greater privacy, and predictable governance.

Q32. Can blockchain data contain personal information?

A. It can, but placing sensitive personal data directly on a blockchain creates privacy and compliance risks.

C. Bitcoin, Ethereum, and Ether

Q33. What is Bitcoin?

A. A cryptocurrency and blockchain network designed for peer-to-peer digital payments.

Q34. What is Ethereum?

A. A blockchain platform that supports programmable smart contracts and decentralized applications.

Q35. What is Ether?

A. The native cryptocurrency of the Ethereum network, represented by ETH.

Q36. What is the difference between Bitcoin and Ethereum?

A. Bitcoin primarily focuses on peer-to-peer digital money, while Ethereum also supports general-purpose smart contracts and decentralized applications.

Q37. What is a smart contract?

A. A program deployed on a blockchain that executes according to its coded rules.

Q38. Does a smart contract necessarily require a human intermediary?

A. No. Its programmed operations can execute through blockchain transactions without a traditional intermediary.

Q39. What is a decentralized application (DApp)?

A. An application that uses blockchain or smart contracts for some of its functionality.

Q40. What is gas in Ethereum?

A. A measure of the computational work required to execute operations on Ethereum.

Q41. Why do Ethereum transactions require gas?

A. Gas accounts for computational resources and helps prevent unlimited or abusive execution.

Q42. Who pays the transaction fee?

A. Typically, the account submitting the transaction pays the applicable fee.

Q43. What is Wei?

A. A very small denomination of Ether.

Q44. How many Wei equal one ETH?

A. \(1\text{ ETH}=10^{18}\text{ Wei}\).

Q45. What is a wallet?

A. Software or hardware that manages blockchain keys and helps users interact with their accounts.

Q46. Does a crypto wallet literally store coins inside it?

A. No. It manages keys that allow users to access and authorize transactions associated with blockchain records.

D. MetaMask and Ether transactions

Q47. What is MetaMask?

A. A cryptocurrency wallet that lets users manage Ethereum-compatible accounts and connect to decentralized applications.

Q48. Why do we install MetaMask?

A. To create or import an account, manage keys, view balances, and approve transactions or DApp interactions.

Q49. What is a seed phrase?

A. A sequence of words used to recover a wallet or derive its keys.

Q50. Should we share our seed phrase with anyone?

A. No. Anyone who obtains it may gain control of the wallet.

Q51. What is a wallet address?

A. A public identifier used to receive cryptocurrency or interact with a blockchain account.

Q52. What is the difference between an address and a private key?

A. An address can be shared to receive funds; the private key authorizes transactions and must remain secret.

Q53. What information can be checked for a transaction?

A. The transaction hash, sender, recipient, amount, fee, block information, and confirmation status, depending on the explorer.

Q54. What is a transaction hash?

A. A unique identifier used to look up a blockchain transaction.

Q55. What is a blockchain explorer?

A. A tool or website for inspecting blocks, addresses, transactions, and network activity.

Q56. What does a pending transaction mean?

A. The transaction has been submitted but has not yet been finalized or confirmed as expected.

Q57. Can a confirmed blockchain transaction normally be reversed?

A. Not through an ordinary reversal mechanism. A separate transaction may be needed to return funds, if the recipient cooperates.

Q58. What is a testnet?

A. A blockchain network intended for testing, generally using test assets rather than real-value mainnet assets.

Q59. Why use a testnet in a practical?

A. To learn how to create transactions and deploy contracts without risking real funds.

Q60. How can you check Ether spent per transaction?

A. Inspect the transaction details and calculate the fee from the gas used and the effective gas price. For a simple ETH transfer, distinguish the amount sent from the transaction fee.

Q61. What is the difference between transaction value and transaction fee?

A. Transaction value is the asset being transferred; the fee pays for processing the transaction.

E. Solidity and smart contracts

Q62. What is Solidity?

A. A programming language commonly used to write smart contracts for Ethereum-compatible blockchains.

Q63. What is a Solidity contract?

A. A program containing state variables, functions, and other declarations that define on-chain behaviour.

Q64. What is a state variable?

A. A variable declared at contract level whose value can persist in blockchain state.

Q65. What does public mean in Solidity?

A. It specifies that a function or state variable has public visibility, subject to the language's rules.

Q66. What does require() do?

A. It checks a condition and reverts execution if the condition is false.

Q67. What does msg.sender represent?

A. The address that directly called the current function.

Q68. What is a mapping?

A. A key-value data structure commonly used to associate addresses with values, such as account balances.

Q69. What is a payable function?

A. A function that is permitted to receive Ether.

Q70. What is a constructor?

A. A special function executed when a contract is created, typically to initialize its state.

Q71. What does deploying a smart contract mean?

A. Submitting a contract-creation transaction so the contract becomes available at an address on the blockchain.

Q72. What is Remix IDE?

A. A browser-based development environment commonly used to write, compile, deploy, and test Solidity contracts.

Q73. What is the difference between compiling and deploying?

A. Compiling converts source code into contract artifacts; deploying submits a transaction to create the contract on a network.

Q74. What is reentrancy?

A. A vulnerability in which an external call allows execution to re-enter a contract before the original operation has safely completed.

Q75. How can reentrancy risk be reduced?

A. Follow checks-effects-interactions, update state before external calls, and use suitable security protections.

Q76. What happens if a require() condition fails?

A. The transaction or call reverts, and state changes made by that execution are rolled back.

Q77. Can smart contracts access ordinary websites directly?

A. Not directly in the usual execution model. External data is commonly supplied through oracles.

Q78. What is an oracle?

A. A service or mechanism that provides external information to a smart contract.

F. Hyperledger and business networks

Q79. What is Hyperledger?

A. An open-source umbrella project for enterprise blockchain technologies.

Q80. What is Hyperledger Fabric?

A. A permissioned distributed-ledger platform designed for enterprise use.

Q81. How does a permissioned blockchain differ from a public blockchain?

A. Participation and access are controlled by defined identities and permissions.

Q82. What is a business network?

A. A network in which participants exchange assets or execute transactions according to shared rules.

Q83. What is an asset in a business network?

A. Something of value represented in the system, such as a product, property, or record.

Q84. What is a participant?

A. An entity or member that can interact with the business network.

Q85. What is a transaction in Hyperledger?

A. An operation submitted by a participant that changes or interacts with the network's state according to its rules.

Q86. What is a connection profile?

A. Configuration information used by a client to connect to a blockchain network.

Q87. What is an identity in a permissioned network?

A. Information and credentials used to identify a participant or application and determine its permissions.

Q88. What is the purpose of a query?

A. To retrieve information from the network's state.

Q89. What is an event?

A. A notification emitted by the network or application when a defined action or state change occurs.

Q90. Give one real-world use case of Hyperledger.

A. Supply-chain tracking between manufacturers, distributors, and retailers.

Q91. Why might an enterprise prefer a permissioned blockchain?

A. It needs controlled participation, defined governance, and restricted access to business data.

G. Comparison and tricky questions

Q92. Is every distributed ledger a blockchain?

A. No. Blockchain is one type of distributed-ledger structure.

Q93. Is every blockchain public?

A. No. Blockchains may be public, private, hybrid, or consortium-based.

Q94. Does hashing alone encrypt data?

A. No. Hashing produces a digest; it is not the same as encryption.

Q95. What is the difference between hashing and encryption?

A. Hashing is generally one-way; encryption is designed to be reversible with the appropriate key.

Q96. Why is a previous block's hash stored in a block?

A. To link the chain and make changes to earlier records detectable.

Q97. What is Proof of Work?

A. A consensus mechanism in which participants perform computational work to propose blocks.

Q98. What is Proof of Stake?

A. A consensus mechanism in which validators participate based on staked assets and protocol rules.

Q99. Is Proof of Work the same as Proof of Stake?

A. No. They use different mechanisms to select and incentivize participants and secure consensus.

Q100. What is a consensus mechanism's purpose?

A. To help participants agree on the accepted state of the blockchain.

Q101. Why is blockchain useful when multiple organizations do not fully trust one another?

A. It can provide a shared, auditable record with common validation rules and less dependence on one central recordkeeper.

Q102. What is one limitation of blockchain?

A. Scalability, transaction costs, privacy, governance, and irreversible mistakes can be significant limitations.

Q103. Can blockchain guarantee that stored information is true?

A. No. It can help preserve recorded information, but incorrect information can still be entered.

Q104. Why is key management important?

A. Losing or exposing private keys can result in permanent loss of control over the associated assets.

Q105. How is blockchain different from a traditional database?

A. A traditional database is usually controlled by a central administrator; a blockchain coordinates shared records through a network protocol and consensus rules.

Extra viva questions that can be asked in any group

These are the questions examiners often ask after you explain an algorithm or demonstrate your practical.

A. Programming and code logic

Q1. Which programming language did you use?

A. Python, because it provides simple syntax and libraries for implementing algorithms and machine-learning models.

Q2. What is a function?

A. A reusable block of code that performs a specific task.

Q3. What is a parameter?

A. A variable that receives a value when a function is called.

Q4. What is a return statement?

A. It sends a result back to the caller and ends the function's execution.

Q5. What is a loop?

A. A control structure that repeatedly executes a block of code.

Q6. What is an array or list?

A. A data structure used to store multiple values in a sequence.

Q7. What is a conditional statement?

A. A statement such as if or else that executes code depending on a condition.

Q8. What is debugging?

A. Finding and fixing errors in a program.

Q9. What is an exception?

A. An error condition that occurs during program execution and can be handled using exception-handling mechanisms.

Q10. What is the difference between a syntax error and a logical error?

A. A syntax error violates language rules; a logical error allows execution but produces an incorrect result.

B. Algorithm analysis

Q11. What is an algorithm?

A. A finite sequence of steps to solve a problem.

Q12. What is time complexity?

A. A measure of how an algorithm's running time grows with input size.

Q13. What is space complexity?

A. A measure of how an algorithm's memory requirements grow with input size.

Q14. What is Big O notation?

A. It expresses an asymptotic upper bound on an algorithm's growth rate.

Q15. Name common time complexities.

A. $O(1)$, $O(\log n)$, $O(n)$, $O(n\log n)$, $O(n^2)$, and exponential time.

Q16. What is the best case?

A. The most favourable input condition for an algorithm.

Q17. What is the worst case?

A. The input condition that produces the greatest running time for a given input size.

Q18. What is average-case complexity?

A. The expected running time over a specified distribution of inputs.

Q19. What is the difference between time and space complexity?

A. Time complexity measures growth in computational work; space complexity measures growth in memory usage.

Q20. Can two algorithms solve the same problem with different complexities?

A. Yes. Different approaches can have different running times and memory requirements.